

ORIGINAL ARTICLE

Integrated Food Science

Assessment of Optical Properties and Monte Carlo Multilayered Simulation of Light Propagation in Cherries for SSC and MC Detection

Jian Wu  | Xianhuan Zeng | Aiguo Ouyang | Bin Li  | Nan Chen  | Yande Liu

National and Local Joint Engineering Research Center of Fruit Intelligent Photoelectric Detection Technology and Equipment, East China Jiaotong University, Nanchang, PR China

Correspondence: Jian Wu (wujian@ecjtu.edu.cn) | Yande Liu (2607@ecjtu.edu.cn)**Received:** 15 September 2025 | **Revised:** 18 December 2025 | **Accepted:** 7 January 2026**Keywords:** cherry tissues | integrating sphere | optical properties | Monte Carlo simulation | internal qualities detection**ABSTRACT**

Nondestructive detection technology for cherry quality is a critical demand in the field of agricultural product processing, whereas existing studies lack a systematic exploration of the optical properties of cherries. This study investigated the optical properties (OPs) of cherry tissues and the light propagation using Monte Carlo multilayered (MCML) simulations. Moreover, the relationships between the OPs with internal qualities were explored. The results showed that skin tissues had significant absorption in 400–600 nm, while decreasing in 650–1050 nm. The reduced scattering coefficient of flesh tissues has significant differences in 400–1050 nm. Microscopic images revealed that the differences in OPs were caused by the broken cellular structure, and the bruising had no significant effect on soluble solids content (SSC) and moisture content (MC). MCML simulations indicated that photon energy distribution was dominated by diffuse reflectance due to the high reflectance of the kernel. Therefore, the reflectance model is more effective for detecting internal qualities. The PLSR models for the prediction of SSC and MC built on the absorption coefficient spectra had the best prediction performance, with a coefficient of determination of 0.935 and 0.839, and root mean square error of 0.254 and 0.008, respectively. The results indicate that the absorption characteristics of cherry tissue are primarily influenced by SSC and MC, and the use of the absorption coefficient enables more accurate prediction of its internal quality. This work offers guidance for nondestructive assessment methods targeting cherry internal quality.

Practical Applications

The reflectance model is more effective for detecting internal qualities, and the internal quality can be predicted better based on the absorption coefficient. This work offers guidance for nondestructive assessment methods targeting cherry internal quality.

1 | Introduction

Chilean cherries are known for their vibrant color and taste, popular among consumers worldwide for abundant health benefits, which implemented a record area of 63,500 hectares planted in 2023 and an annual production of about 470,000 tons (FAOSTAT 2023). As consumer demand for fruit quality continues to rise, the

development and application of efficient, nondestructive cherry quality assessment technologies have become increasingly critical. In recent years, visible near-infrared (VIS-NIR) spectroscopy has been widely used for the nondestructive measurement of several chemical constituents, such as moisture, soluble solids, and acidity (Guo et al. 2024; Wei et al. 2020; Xie et al. 2021). VIS-NIR spectroscopy acquired outstanding classification perfor-

mance by exploiting the sensitivity of fruit tissue components to specific wavelengths. However, VIS-NIR detection models usually use statistical black-box methods to relate the spectra to chemical/physical properties of fruits, but the internal scattering and complex light propagation were weakened or ignored, which limited the applicability of the models to training conditions (Cho et al. 2022). Separating the optical properties of cherry tissues from other interfering factors is expected to provide more reliable classification results for fruit quality under variable conditions.

To effectively improve the application of VIS-NIR spectroscopy for detecting cherries' internal qualities, it is crucial to understand the optical properties of cherry tissues and light propagation. Cherry tissues are considered turbid media, where light interaction involves complex processes of absorption and scattering due to the chemical composition and physical structure (Y. Guo et al. 2025). The interaction results in light attenuation, influenced by the combined effects of tissue absorption and scattering (Wei et al. 2020; Wang et al. 2025b). Therefore, VIS/NIR interaction with cherry tissues produces spectra that reflect the layer-specific optical properties. These optical properties are primarily described by absorption coefficients (μ_a), which are related to chemical components (sugars, chlorophyll, soluble solids, and water; Xie et al. 2021), and scattering coefficients (μ'_s) affected by structural attributes (density, particle size, refractive index, and cellular structure) (Jiang et al. 2023). Hence, tissue optical properties offer key insights into chemical composition and structural features, supporting nondestructive evaluation of fruit quality (Hu et al. 2024). Additionally, optical properties include the anisotropy factor (g) and refractive index (n). The g refers to the average cosine of the scattering angle of photons when they are scattered in the flesh tissue. The n is defined as the ratio of the speed of light in a vacuum (c) to that in a medium (v). The $\mu'_s(\mu'_s = \mu_s(1-g))$, derived by combining the g and the μ_s generally, which further characterizes how light propagates through tissues (Zhang et al. 2019). The assessment of the optical properties is helpful to develop more effective VIS-NIR spectroscopy methods for assessing the quality of cherries and other fruits.

Multiple measurement techniques of the optical properties of biological tissue have been widely used for quality assessment of food and agricultural products. Sun et al. (2022) quantitatively characterized the difference of the μ_a and μ'_s values between bruised and healthy apples by spatial frequency domain imaging (SFDI), confirming the sensitivity of optical properties to early bruising. Sun, Lu, and Wang (2020) used spatially resolved (SR) diffuse reflectance spectroscopy to acquire the optical parameters of healthy and fungus-infested peach fruit tissue, and the microstructure and physicochemical properties were also measured. The results demonstrate that optical properties enable effective early detection of peach disease infection. Ma et al. (2023) used time-resolved (TR) transmission spectroscopy to reveal the dynamic decay pattern of optical properties during storage, and found that the μ'_s values decreased with storage time. Wei et al. (2020) employed an integrating sphere (IS) system to research the relationship of the optical properties with the internal quality of apples, finding the μ_a and μ'_s as well as the contents decreased with storage time. Although the detection techniques of optical properties are commonly applied to assess fruit quality, the OPs of cherry skin and flesh tissues with dark

color remain underexplored. Furthermore, the impact of bruising on cherry internal quality, microstructure, and OPs remains insufficiently understood. Assessing OPs in cherry tissues is necessary to decipher how OPs relate to tissue structure and internal quality.

Visible/near-infrared spectral analysis of cherries typically employs transmittance and reflectance measurements (Elsherbiny et al. 2025). To maximize the acquisition of chemical composition information from the fruit pulp, it is essential to comprehensively analyze the propagation characteristics of light through multiple tissue layers and clearly characterize the optical properties of each layer. Monte Carlo multilayered (MCML) simulation as a probabilistic approach to describe the propagation of photons in biological tissues based on the estimation of μ_a and μ'_s , that is considered to be the standard method for exploring the complex light propagation in tissues (Yang et al. 2025). In addition, the finite element method (FEM), formulated within the diffusion approximation (DA), is among the computational tools used to study light propagation in turbid media (C. Guo et al. 2025). Conventional MCML algorithms have been proven to accurately characterize the photonic interaction across multilayered biological tissues. For example, Zhang et al. (2019) revealed the light propagation characteristics of photons in three-layer tissues of bruised blueberry fruits by Monte Carlo multilayered simulation. To improve the biological suitability of the light propagation model, Fang and Yan (2022) creatively constructed a voxel grid-based computational architecture and developed an MC simulation program, such as the Monte Carlo eXtreme (MCX) algorithm. M. Wang et al. (2024) applied MCX to model light propagation in the flavedo of citrus fruits infected by *Penicillium*. More complex simulations can also be achieved through the MCX method. Tang and Yao (2021) achieved the simulation of light transmission in the mouse cerebrovascular system. Although MCML methods have been widely used in the study of light propagation within fruits, most studies have not considered the influence of the fruit kernel. The study of the influence of fruit kernels on the light propagation path can provide the theoretical basis for selecting a suitable spectral detection model.

This study aimed to evaluate the optical properties (OPs) of cherry tissues and model light propagation using MCML simulation, and to examine associations with internal qualities—soluble solids content (SSC) and moisture content (MC). Specific objectives include: (1) building and calibrating the single integrating sphere system to measure the total transmittance and reflectance spectra of cherries flesh and skin; (2) estimating the optical properties of the cherry tissue layers (skin and flesh) by the IAD algorithm; (3) studying the light propagation in three-layer tissues (skin, flesh and kernel) of cherry fruit by MCML simulation; (4) analyzing how bruising alters cherry internal quality and microstructure, and investigated associations between tissue OPs and SSC and MC.

2 | Materials and Method

2.1 | Sample Preparation

Chilean cherry variety 'Santina' was used in this study, which has a round or heart-shaped fruit with a smooth purple-black

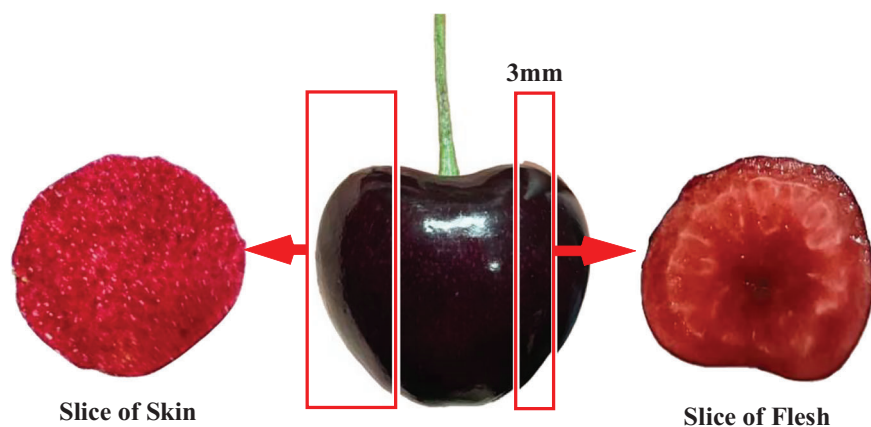


FIGURE 1 | Preparation of cherry flesh and skin samples.

TABLE 1 | Sample size of treatments.

Treatment	Healthy	Fully bruised			Total
		1 h	24 h	48 h	
Flesh	15	15	15	15	60
Skin	15				15
Total	30	15	15	15	75

skin and crisp flesh, and consists of pedicel, skin, flesh, and kernel (Figure 1). The samples were purchased in November 2024 from the Shenzhen agricultural products wholesale market, Nanchang, Jiangxi Province, China. Before the experiment, the samples were kept at $20 \pm 1^\circ\text{C}$ and $90 \pm 5\%$ RH for 24 h to reach room temperature in the laboratory, to eliminate extrusion deformation and temperature differences. The experiment used 75 cherries that were uniform in size and free from apparent deformities.

The cherry samples were divided into four groups; one group consisted of 30 healthy (nonbruised) samples, while the other three groups were artificially bruised with 15 samples in each treatment (Table 1). Each cherry in the bruised groups was dropped freely from a height of 0.5 m onto a plastic plate 4–5 times, to induce thorough and evenly distributed bruises on the flesh of cherries. After dropping, the cherries were inspected to confirm that the drop height would not cause skin rupture, leading to juice oozing. The bruised samples were kept at room temperature for 1, 24, and 48 h to generate distinct bruising levels.

The methods for sample handling and preparation of cherry fruits are illustrated in Figure 1. Firstly, the cherries were initially sliced horizontally to remove a 2–3 mm outer layer. Then, the cherries were cut into thin slices with a thickness of 3 mm, using a razor blade and custom spacers. The flesh tissue slices were sandwiched between two microscope slides to minimize oxidative browning by limiting air exposure, preserving the integrity of the samples for more accurate measurement. The measurement time was kept 15–20 min to ensure the optical properties were stable. Skin samples were selected from the equatorial portion of the fruit (at the largest cross-section of the fruit), a region that is usually

uniformly developed and easy to locate, reducing bias due to differences in fruit morphology. When collecting skin tissues, large pieces of cherry fruit were cut along the pedicel-kernel, and the peel tissue from the large cherry pieces were extract. The diameter of the skin tissue should be greater than 12.8 mm. The skin tissues were placed between two microscope slides and held in place for subsequent measurements. The combined thickness of the slides and skin tissue was determined with a spiral micrometer. Skin thickness was calculated as the difference between the total measurement and the slide thickness.

2.2 | Measurement of Internal Quality and Microstructure

The SSC of each sample was measured using a digital refractometer (PAL-1, Atago Co., Ltd., Tokyo, Japan) on 0.2 mL of cherry juice extracted from the cherry flesh. About 1–2 g of cherry flesh was dried using an electric thermostatic blast drying oven (DHG-101-1SA, Shanghai Hongdu Electronic Technology Co., Ltd., China) until the weight remained stable. Moisture content was determined by measuring mass loss during drying. All analyses were performed in triplicate, and the mean and standard error were calculated from the replicate data.

The microstructure of cherry flesh tissues was investigated by an inverted fluorescence microscope. Cherry flesh tissues were cut into thin slices of 0.5 mm thickness using a razor blade and customized spacers. The slices were subsequently visualized using an inverted fluorescence microscope (37XE-PC, Shanghai optical, Shanghai, China) with a magnification of 40 $\times$, and images were captured using Image View software.

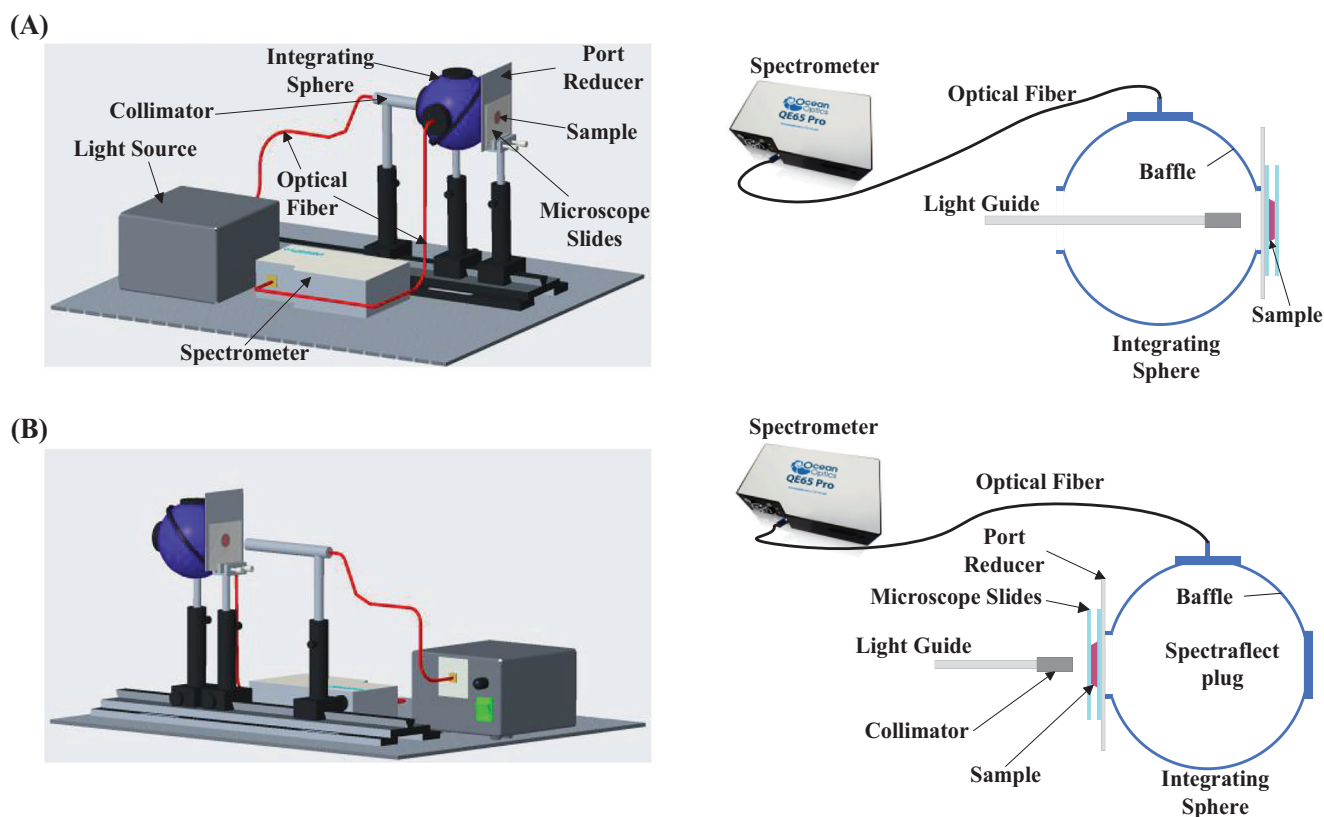


FIGURE 2 | Schematic representation of measurement patterns: (A) total reflectance; (B) total transmittance.

2.3 | Integrating Sphere System Setup

A single integrating sphere (SIS) system was implemented for acquiring cherry tissue spectra in this study. The SIS system (Figure 2) included a master computer, integrating sphere (4P-GPS-033-SL, Lab sphere, USA), spectrometer (QE65pro, Ocean Optics, USA), halogen light source (HL-2000-FHSA, Ocean Optics, Inc.), fiber optics, collimator mirror (74-UV, Guangzhou Hangxin Optoelectronics Technology Co., Ltd., China), adapters, and slide unit. The main body of the integrating sphere is a sphere with an inner wall coated with a PTFE diffuse reflective coating with a reflectivity of >98%. Two diameter ports are symmetrically distributed at the equatorial position of the sphere to serve as the optical input and output interfaces, respectively. Light-reflective-coated port covers were used to seal the integrating sphere ports.

The light beam is emitted from the light source and transmitted to the integrating sphere port through the 800 μm core diameter optical fiber of the SMA905 interface, and then through the 74-UV collimator mirror. The QE65pro spectrometer is configured with a wide spectral response range of 347.1–1171.7 nm. The effective analytical wavelength range is set to 400–1050 nm, avoiding the interference of detector noise at the UV end and the influence of water molecule absorption bands at the near infrared end. The adapter assembly is made of custom aluminum (outside diameter of 15 mm, inside diameter of 13 mm, length of 140 mm), which is precision-machined to ensure the coaxial alignment of the optical fiber with the collimator mirror. The length was designed to be greater than the integrating sphere diameter to effectively reduce the influence of stray light.

The ports of the integrating sphere are difficult to directly adapt to the sliced samples due to the morphological characteristics of the cherry fruit. Therefore, an aluminum port reducer with geometric dimensions of 75 mm $\times$ 100 mm $\times$ 1 mm (central aperture diameter of 12.8 mm) was used as the optical interface component in this study. The port reducer is precisely machined to optimize the geometrical parameters of the port and is placed between the sample-integrating sphere ports, which effectively solves the problem of mechanical adaptation of small-sized samples to the standard detection system.

2.4 | Measurement of Cherry Tissue Optical Properties

The SIS system requires a 30 min thermal equilibration process at start-up to ensure a stable photodetector response. The reference spectrum and dark spectrum were collected to accurately calibrate the system and to mitigate the potential impact of system noise or external interference. First, the integrating sphere port is covered with a reflective port cap (Figure 2A). The light guide is inserted into the sphere to maintain a constant distance between the collimator and the sample. The light source is turned off to measure the dark reflectance spectrum (R_d), and then turned on to obtain the bright reference spectrum (R_r).

The transmittance reference spectrum was measured by the method shown in Figure 2B. The light source was turned off, and the sphere outlet was then covered by a port cap. The port reducer was placed between the light guide and the input port of the integrating sphere for measuring the dark transmittance spec-

TABLE 2 | The input parameters for the IAD program.

Parameter type	IAD input parameters	Values
Basic parameters	Index of refraction of the flesh sample	1.35
	Index of refraction of the skin sample	1.365
	Index of refraction of the top and bottom slides	1.5
	Thickness of samples (mm)	3
	Thickness of slides (mm)	1
	Diameter of illumination beam (mm)	3
Parameters of integrating sphere	Sphere diameter (mm)	83.82
	Input port diameter (mm)	25.4
	Output port diameter (mm)	25.4
	Port reducer diameter (mm)	12.8
	Detector port distance (mm)	3
	Reflectivity of the sphere wall	0.98

trum (T_d). Then, the light source was turned on to measure the bright transmittance reference spectrum (Tr). During reflectance (R_s) and transmittance (T_s) measurements, the collimator-sample distance was set to 3 mm to produce a 3 mm-diameter spot. The transmission (T) and reflection (R) spectra of the sample were obtained using Equations (1) and (2).

$$R = \frac{Rs - Rd}{Rr - Rd}. \quad (1)$$

$$T = \frac{T_s - T_d}{Tr - T_d}. \quad (2)$$

All single integrating sphere system experiments were conducted in a darkroom to eliminate ambient light interference. For each sample, the measurements were repeated three times independently to ensure repeatability, and the average value was adopted as the final result. The reflectance and transmittance spectra for each sample were saved in a single file, and then the inverse adding-doubling (IAD) algorithm was applied to determine μ_a and μ'_s . The refractive index measurement method was described in detail by Zhang et al. (2019). Table 2 summarizes the IAD program input parameters.

2.5 | System Validation and Estimation of Optical Properties

The SIS system needs to validate the accuracy before the experiment. Pure water was utilized as an absorbing agent due to its stability and minimal temperature sensitivity. Meanwhile, Intralipid-20% fat emulsion (medium/long-chain fat emulsion injection C8-24, Anhui Fengyuan Pharmaceutical Co., Ltd., China) was used as the standard scattering medium, which has the characteristics of good monodispersed and homogeneous particle size distribution. To ensure the reliability of the measurement system, the μ_a of pure water and the μ'_s of the 1% intralipid solution were measured. Then the measured values were compared with the results of the literature to confirm the accuracy of the SIS system. The 1% intralipid solution was contained in a cuvette with a thickness of 1 mm. The pure water

was measured using a cuvette with a thickness of 5 mm to enhance the photon absorption. Finally, the average relative error was calculated using Equation (3):

$$R_{error} = \frac{|\mu_{measure} - \mu_{reference}|}{\mu_{reference}} \times 100\%, \quad (3)$$

where $\mu_{measure}$ is the measured value, $\mu_{reference}$ is the reference value, and R_{error} represents the relative error.

Photon migration in biological media is nonlinearly modulated by the absorption-scattering coupling; consequently, light attenuation is commonly characterized by the penetration depth. Penetration depth (σ) is the effective distance at which the diffuse light intensity falls to $1/e$ ($\sim 36.8\%$) of the initial irradiance along the tissue depth direction (Xiao et al. 2023). The σ were calculated by the diffusion approximation Equation (4):

$$\sigma = \frac{1}{\sqrt{3\mu_a(\mu_a + \mu'_s)}}. \quad (4)$$

2.6 | Monte Carlo Multilayered Simulation

Based on the spatially constrained properties of light propagation inside fruits, the multilayered turbid medium model was used in this study. The skin, flesh, and kernel layers of the cherry were considered as homogeneous media, and the tissue interfaces were set to ideal optical flatness (Ding et al. 2015). The strong attenuation properties of the kernel force the spectral system to use high-intensity incident light, which is prone to triggering photothermal skin bruising, leading to tissue browning, when adopting the transmittance mode (Rahman et al. 2025). Therefore, this study focuses on the optical behavior of the kernel in reflectance mode. The incident photons are reflected by the kernel interface, resulting in a secondary optical path, after penetrating the flesh medium. In reflectance mode, part of the incident light is reflected by the kernel and reabsorbed/scattered in the flesh. Figure 3 shows the interaction of light with a simplified model of cherry tissue. Monte Carlo radiative transfer models

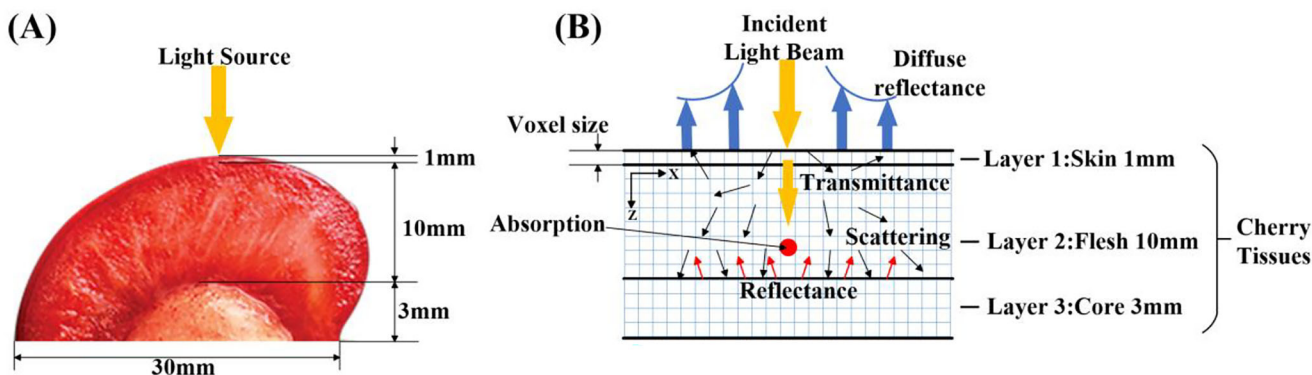


FIGURE 3 | MCML simulation of light interaction with cherry tissue: (A) view of cherry profile; (B) interaction of light with simplified model of cherry tissue.

TABLE 3 | Input parameters for MC simulation of light propagation in cherry tissue.

Parameter	Value
Number of photons	100,000,000
Voxel size	1
Number of layers	3
Refractive index of medium above and below tissue	1
Refractive index of 1st layer, 2nd layer, and 3rd layer	1.365; 1.35; 1.46
The scattering anisotropy factor (g)	0.7

were constructed to quantitatively describe the contribution of the kernel to the overall light propagation, including a two-layer (skin-flesh) model and a three-layer (skin-flesh-kernel) model. The MCX program (Fang and Yan 2022) was used to calculate path lengths and light attenuation of detected photons for studying the light propagation in multilayered cherry tissues, which employs a microscopic approach based on the Beer-Lambert law (Hirvi et al. 2023). Following this strategy, optical attenuation is governed by μ_a , whereas photon propagation paths are dictated by μ'_s .

In this study, 10 million photons were used to ensure more accurate results. A voxel size of 1 millimeter was employed, with layer thickness values derived from the average of 15 cherry samples. The OPs and refractive indices were both determined based on prior experimental measurements. The value range of g for biological tissues is 0.4–0.9, and studies have found that the optical properties are not sensitive to the g value, which was determined to be 0.7 for this study (Ma et al. 2024). The necessary input parameters for the MCX simulation are listed in Table 3.

2.7 | Data Analysis

To investigate the relationship between the OPs of cherries and their internal quality content, chemometric methods were employed for sample division and modeling. A joint X–Y distance (SPXY) algorithm allocated samples to a calibration set (90

samples) and a prediction set (30 samples) at a 3:1 ratio. Separate partial least squares regression (PLSR) models for soluble solids content (SSC) and moisture content (MC) were developed using μ_a and μ'_s , respectively. The entire process was conducted in The Unscrambler X 10.4 (The Como Inc., Norway). Calibration was evaluated with the coefficient of determination for calibration set (R^2_c) and the root mean square error of calibration (RMSEC); prediction with the coefficient of determination for prediction set (R^2_p) and the root mean square error of prediction (RMSEP).

3 | Results and Discussion

3.1 | Validation of the System

The reference values of μ_a for pure water were obtained in the report of Deng et al. (2012), and the μ'_s reference of Intralipid-20% fat emulsion injections was calculated by the empirical formulas given by Michels et al. (2008) (Equations 5 and 6):

$$\mu'_s(\lambda) = 77.23 - 0.1131\lambda + 5.122 \times 10^{-5}\lambda^2, \quad (5)$$

$$\mu'_s(\rho_{il}) = \mu'_s(\lambda) \rho_{il}, \quad (6)$$

where λ is the wavelength, $\mu'_s(\lambda)$ represents the μ'_s values of Intralipid-20% fat emulsion at λ wavelengths, and ρ_{il} represents the volumetric concentration of the solvent.

Measurements of μ_a (pure water) and μ'_s (Intralipid-20%) obtained with the SIS system in the 400–1050 nm range agreed well with the reference data (Figure 4). The curve of μ_a values (Figure 4A) have a prominent O–H bond absorption peak at 980 nm and shows a high degree of similarity in the range of 800–1050 nm, with the average error of 6.91%. Studies by Ma et al. (2020) and Wei et al. (2020) also identified a noticeable baseline shift in the 400–800 nm spectrum. The shift may result from the loss of direct and diffused light at the sample edges, leading to an elevated absorption coefficient. The measured μ'_s values (Figure 4B) show the Mie scattering characteristic, and decrease with increasing wavelength. The deviation of the measured values from the reference values may be caused by the different particle size distribution characteristics of the different fat emulsion suppliers. Wei et al. (Wei et al. 2020) reported that 1%

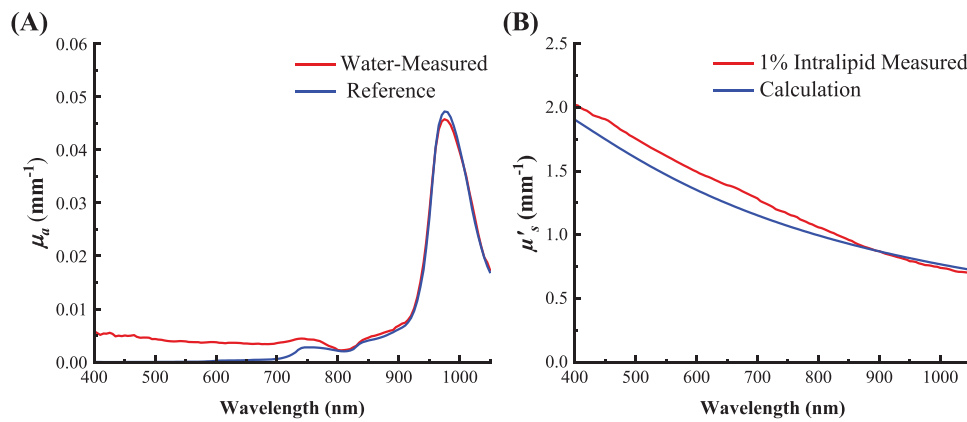


FIGURE 4 | (A) The μ_a of pure water and (B) the μ'_s of 1%-Intralipid at 400–1050 nm.

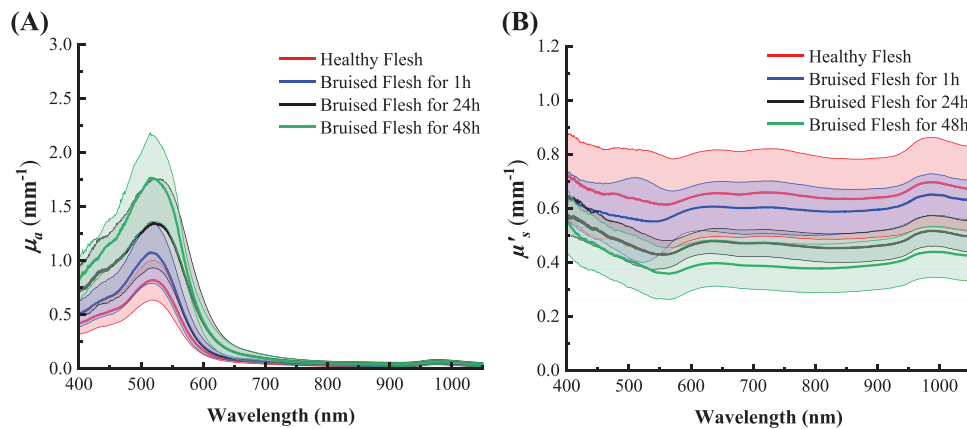


FIGURE 5 | Comparison of the (A) μ_a and (B) μ'_s of the healthy, 1 h-bruised, 24 h-bruised, and 48 h-bruised flesh tissues.

fat emulsions tended to be overestimated between 800 and 830 nm and underestimated above 830 nm; the results are consistent with this study. The measured μ'_s exhibited relative errors of 0.22%–12.29% (average 7.04%), outperforming the 8.79% reported by Xiao et al. (2023).

3.2 | Optical Properties of Cherry Flesh

The optical properties of the cherry flesh in 400–1050 nm are shown in Figure 5. The average μ_a values of the flesh tissues with different bruising times show a consistent trend with two absorption peaks of different intensities at 520 nm and 980 nm (Figure 5A). The absorption peak near 520 nm can be attributed to anthocyanin. The significant difference in μ_a in 520 nm is associated with browning of the flesh due to mechanical bruising (Sui et al. 2023). The p-coumaric acid is the predominant phenolic acid found in cherry flesh. When the flesh tissue browns, the level of p-coumaric acid decreases sharply, and the polymeric pigment increases rapidly, with the dark pigment having a strong absorption of light. Therefore, the absorption coefficient increases with increasing bruising time. The characteristic absorption peaks in the 980 nm band can be attributed to the octave vibration modes of the O–H bonds in polysaccharides and free water molecules (Vega-Castellote et al. 2023), and μ_a did not differ significantly between healthy and bruised flesh.

The μ'_s values of cherry flesh for four treatments show a significant difference in 400–1050 nm (Figure 5B), and change gently. Due to the destruction of tissue structure, the μ'_s values of bruised flesh are lower than those of nonbruised flesh, and decrease with the increase of bruised time. These results are consistent with those reported by Wang et al. (2025c) on bruised “Kuerle” pears. The μ'_s values are usually significantly correlated with the mechanical response in terms of tissue hardness and modulus of elasticity (Z. Wang et al. 2024; Chen et al. 2025). The significant difference demonstrated that the μ'_s can be used as the core feature for nondestructive identification of cherry bruises. Therefore, the scattering technique is expected to be used in practical applications for the detection of bruised cherries.

3.3 | Optical Properties of Cherry Skin

Figure 6 represents the average μ_a and μ'_s of cherry skin. The μ_a values of the skin are about twice that of the flesh in the spectral range of 400–600 nm (Figure 6A). Anthocyanins have multiple conjugated double bond structures, which give them specific absorption peaks in the visible range, usually around 490–550 nm. Therefore, the absorption spectrum also shows a peak near 520 nm, due to the high content of anthocyanins in the cherry skin. The high absorption coefficient of the skin in

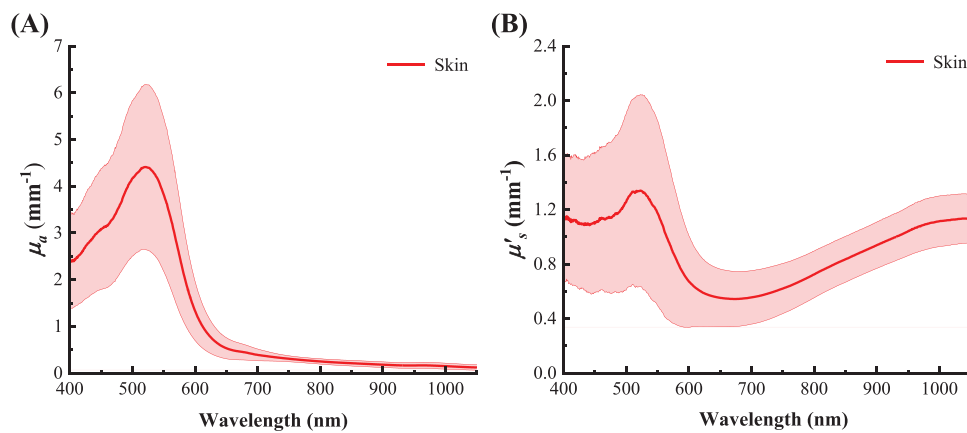


FIGURE 6 | The average spectra (A) μ_a and (B) μ_s of cherry skin tissue.

400–600 nm indicates that photon energy is largely absorbed, resulting in fewer photons entering the pulp. The absorption coefficient becomes lower and flatter in the wavelengths over 600 nm.

The μ_s curves display a peak and valley near 520 and 680 nm (Figure 6B), respectively. The characteristic valley may originate from the coupling of optical energy absorption and Rayleigh scattering processes within the integrating sphere optical system, while the effect of reflected optical path distortion induced by the insertion of the collimator assembly into the integrating sphere cavity needs to be taken into account (Xie et al. 2021). The μ_s of cherry skin does not show a tendency to decrease with increasing wavelength, probably because the size of the basic scattering unit in the skin tissue is already larger than the wavelength.

The μ_a and μ_s values exhibit a large variation in the spectral regions of 400–700 nm. The primary reason may be attributed to the large change of anthocyanin content in cherry skin, resulting in significant fluctuation of optical response in this spectral range. A second factor is that wrinkling of the skin during mounting between two microscope slides can increase the μ_a values, introducing a nonnegligible error. Third, inaccurate skin thickness determination with a spiral micrometer represents an additional source of error. Human errors in sample preparation can lead to variations in the recorded thickness of the skin sample, usually because residual pulp can adhere to the sample.

3.4 | Light Penetration Depth of Cherry Tissues

Light penetration depth was computed from the mean optical properties at each wavelength (Figure 7), which was significantly greater in the flesh layer than in the skin layer. The differences in depth originated from the coupling of the high absorbance properties of the skin layer with multiple scattering mechanisms. However, a favorable light propagation condition was formed in the flesh layer due to low absorption and weak scattering.

The results showed that the light penetration depth increased with bruising time. Healthy flesh showed the lowest light pene-

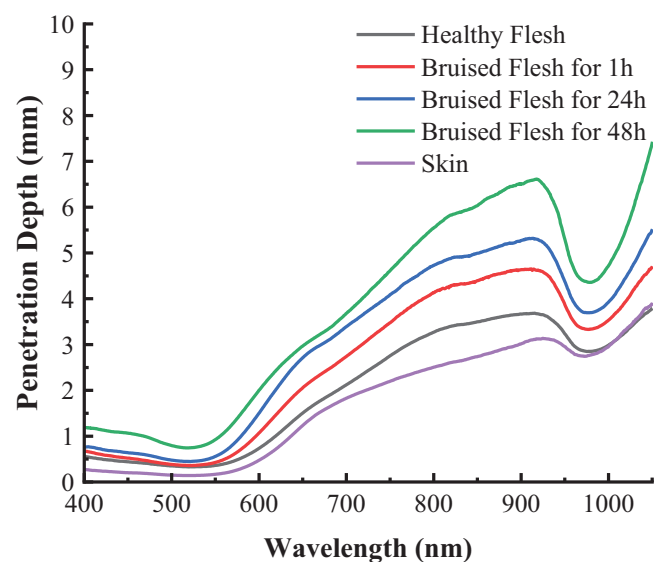


FIGURE 7 | Penetration depth of healthy flesh, bruised flesh (1, 24, and 48 h), and skin samples.

tration depth due to the high density of intact cellular structures. The microstructural changes triggered by bruising enhance light propagation through a dual mechanism: (1) the dissolution of pectin in the intercellular layer reduced the density of scatterers, while cellular rupture led to an increase in the equivalent diameter of scattering particles; (2) the spectral characterization showed that the near-infrared region offers superior deep detection capability compared with the visible region.

In the spectral range of 400–600 nm, the skin had the shortest depth of penetration due to its high absorption and backscattering, and only a very small amount of light penetrated the skin to the flesh. In the 650–1050 nm range, the light penetration depth of the skin was longer, the photon propagation efficiency increased, and the photons interacted with the flesh tissue to provide more information about the bruised flesh. Therefore, the region of 650–1050 nm has greater potential for the detection of bruised cherries.

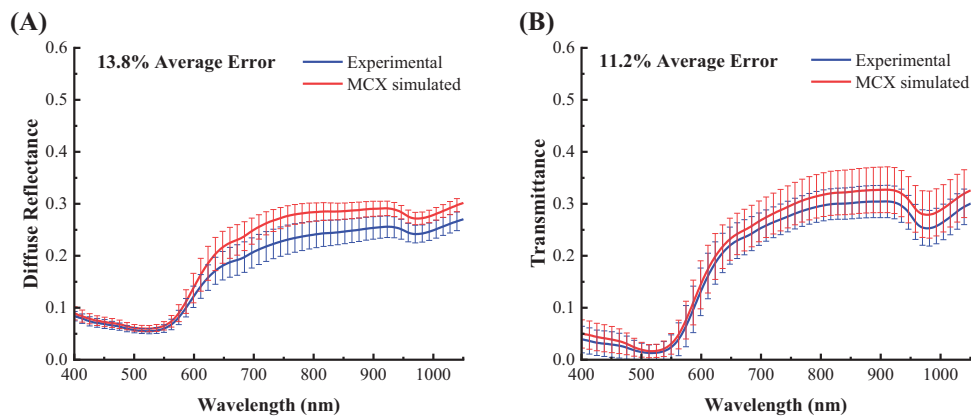


FIGURE 8 | Mean and standard deviation values of experimental data and MCX simulation results for (A) diffuse reflectance and (B) transmittance of healthy flesh samples.

3.5 | Light Propagation Simulations in Multilayered Cherry Tissues

3.5.1 | Monte Carlo Validation Test

The experimental measurements of diffuse reflectance and transmittance were consistently lower than the MCML simulations over the entire wavelength range (Figure 8), with average relative errors of 13.8% and 11.2%. The main sources of error included the sample thickness and the experimental setup, as highlighted by Wang et al. (2025b). In addition, the statistical differences inherent in the IAD algorithm and the MCML simulation process also contributed to these errors (Yang et al. 2024). The results of MCML simulations were close to the experimental data, which confirmed the reliability of the simulation.

3.5.2 | Light Energy Distribution in Multilayered Cherry Tissues

The MC simulation of light interaction with multilayered cherry tissues indicates that photon energy distribution is primarily categorized into diffuse reflection and absorption, with minimal transmission (Figure 9). That confirms the technical bottleneck of the transmittance detection method for the assessment of cherry quality, because the detection sensitivity is limited by the probability of photon escape from deeper tissues. The absorption steadily increases with wavelength, peaking around 520 nm due to chlorophyll absorption, and the increased absorption fraction is observed near 980 nm due to water absorption. Simulation results showed lower diffuse reflectance fraction in the 400–600 nm band and higher values in the 650–1050 nm band, with healthy samples exhibiting significantly higher values than bruised ones. These findings indicate that reflectance measurements in the 650–1050 nm range offer greater potential for detecting bruised cherries.

3.5.3 | Absorption Energy Density Distribution in Multilayered Cherry Tissues

Since the absorption fraction of the multilayered cherry model is maximized at 520 nm, which is also the pigment absorp-

tion wavelength point. In addition, 980 nm is the absorption wavelength point for water. Therefore, 520 and 980 nm were selected as the characteristic wavelengths for light propagation simulation. Figure 10 demonstrates the absorption energy density distribution of different cherry samples at 520 and 980 nm. The depth of absorption (Z-direction) and radius of diffusion (X-direction) of photons at specific wavelengths are further visualized by calculating the contour distribution of the absorbed energy density. As the light propagation increases, the absorbed energy distribution is uniform, and the energy contours are homogeneous, which is in agreement with Yang et al. (2024) and Zhang et al. (2019).

Comparing the absorbed energy distribution at different wavelengths, all samples had a shallower absorption depth at 520 nm (Figure 10A,C). The experimental data (Figure 6A) showed that the skin exhibited a strong light absorption effect ($\mu_a = 4.41 \text{ mm}^{-1}$) in the 520 nm due to anthocyanin absorption, most of the photons undergo energy dissipation in the skin layer. In contrast, the μ_a of the skin was significantly reduced to 0.16 mm^{-1} at 980 nm, and the light penetration depth was increased, resulting in better photon propagation efficiency.

The photons are difficult to penetrate the kernel and instead are reflected due to the special optical properties of the kernel, then reabsorbed and scattered in the flesh layer. Comparing the two models at the same wavelength (Figure 10A,C), there is no significant difference in the depth of absorption and the radius of diffusion at 520 nm. At 980 nm (Figure 10B,F), the three-layer model with a kernel exhibits shallower photon absorption depth and a much larger diffusion radius. The difference arises from the fact that photons in the two-layer model can penetrate tissue layers compared with the three-layer model, in which photons are affected by the fruit kernel. In addition, parts of the photons are reflected to the flesh, resulting in shallower absorption depth and wider diffusion radius. To accurately determine light propagation characteristics, the influence of the kernel in the stone fruits must be considered. The kernel of the cherry occupies a large proportion of the fruit, and the refractive index varies significantly in different tissue layers. Based on the above study, to enhance flesh information capture and maintain high SNR, large detection angles should be avoided as much as possible, and high incident light intensity and high sensitivity detectors are required.

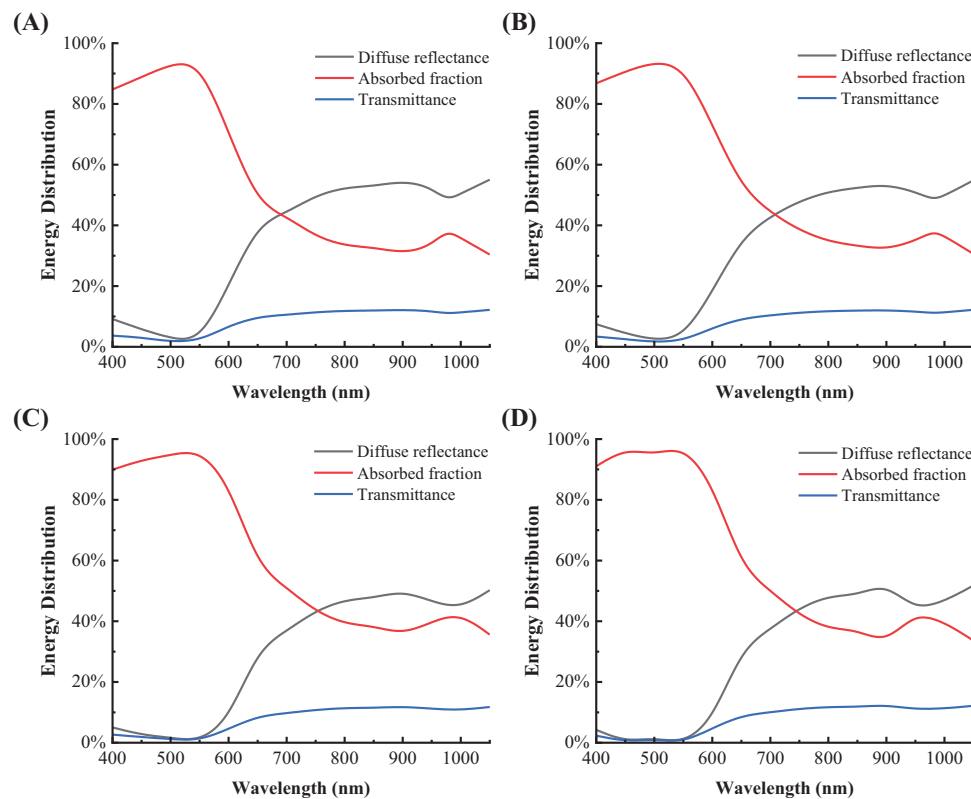


FIGURE 9 | Energy distribution in cherry tissues from different treatments:(A) healthy, (B) 1 h-bruised, (C) 24 h-bruised, and (D) 48 h-bruised flesh.

As shown in Figure 10, the 48 h bruised flesh, with the reduced scattering coefficient ($\mu'_s = 0.44 \text{ mm}^{-1}$), exhibits greater absorption depth and a broader diffusion radius (Wang et al. 2025b). In contrast, as the reduced scattering coefficient of healthy flesh increases ($\mu'_s = 0.7 \text{ mm}^{-1}$), the absorption depth decreases, and the diffusion radius of photon propagation becomes smaller (Figure 10D–F). Tissue layers with larger absorption coefficients absorb photons more rapidly over a smaller area, while those with smaller coefficients allow photons to propagate to deeper areas. Thus, the photon absorption energy distribution at a given wavelength can distinguish the bruised flesh tissue. Reviewing Figures 5 and 6, the light propagation is influenced by Ops, which agrees with the findings of Yang et al. (2024) and Wang et al. (2025a).

3.6 | Evaluation of Bruising Effects on Internal Qualities and Microstructure of Cherries

3.6.1 | Internal Qualities of Cherries' Flesh

The statistical results of cherry internal qualities with different bruising times are presented in Table 4. The increasing of SSC was observed with prolonged bruising duration. The lowest mean SSC value (12.63%) was recorded for healthy cherries, while the highest mean SSC (13.23%) was obtained from 48 h bruised samples. Some differences in SSC were found between healthy and bruised samples. The SSC elevation might be attributed to increased release of sugars and other soluble solids from bruised tissues (Ma et al. 2024).

Regarding moisture content (MC), an initial increase followed by a decrease was observed with extended bruising time. Healthy cherries exhibited an MC of 85.04%, which increased to 86.22% in 24 h bruised samples and subsequently decreased to 85.74% in 48 h bruised samples. This pattern could be explained by cellular structure disruption caused by bruising, leading to leakage of bound water from cells into intercellular free water, resulting in temporary MC elevation during early bruising stages. Subsequently, enhanced respiration and transpiration induced by bruising were considered to cause the MC reduction in 48 h bruised samples (Gao et al. 2021).

3.6.2 | Microstructure of Cherry Flesh and Skin

Inverted fluorescence microimaging was performed on healthy and 24 h-bruised flesh tissues. The microscopic images show the structural differences of the three tissues (nonbruised and 24 h bruised flesh, skin) (Figure 11). The tissue cells of healthy flesh were tightly packed in a polygonal shape, with the epidermal cell layer intact and attached to the flesh matrix (Figure 11A). In contrast, the structure of 24 h bruised flesh tissue showed significant changes (Figure 11B). Mechanical stress led to the disintegration of β -glucan in the intercellular layer, triggering cell wall rupture and extravasation of intracellular fluid, resulting in increased cell dispersion. The observations show that the cell arrangement orderliness index of the bruised tissues decreased, and the cell gap became larger than that of the healthy group, leading to a reduction in the Mie scattering. The skin cells of cherry showed cellular characteristics significantly different from those of the flesh, with the cellular arrangement showing a neat and tight

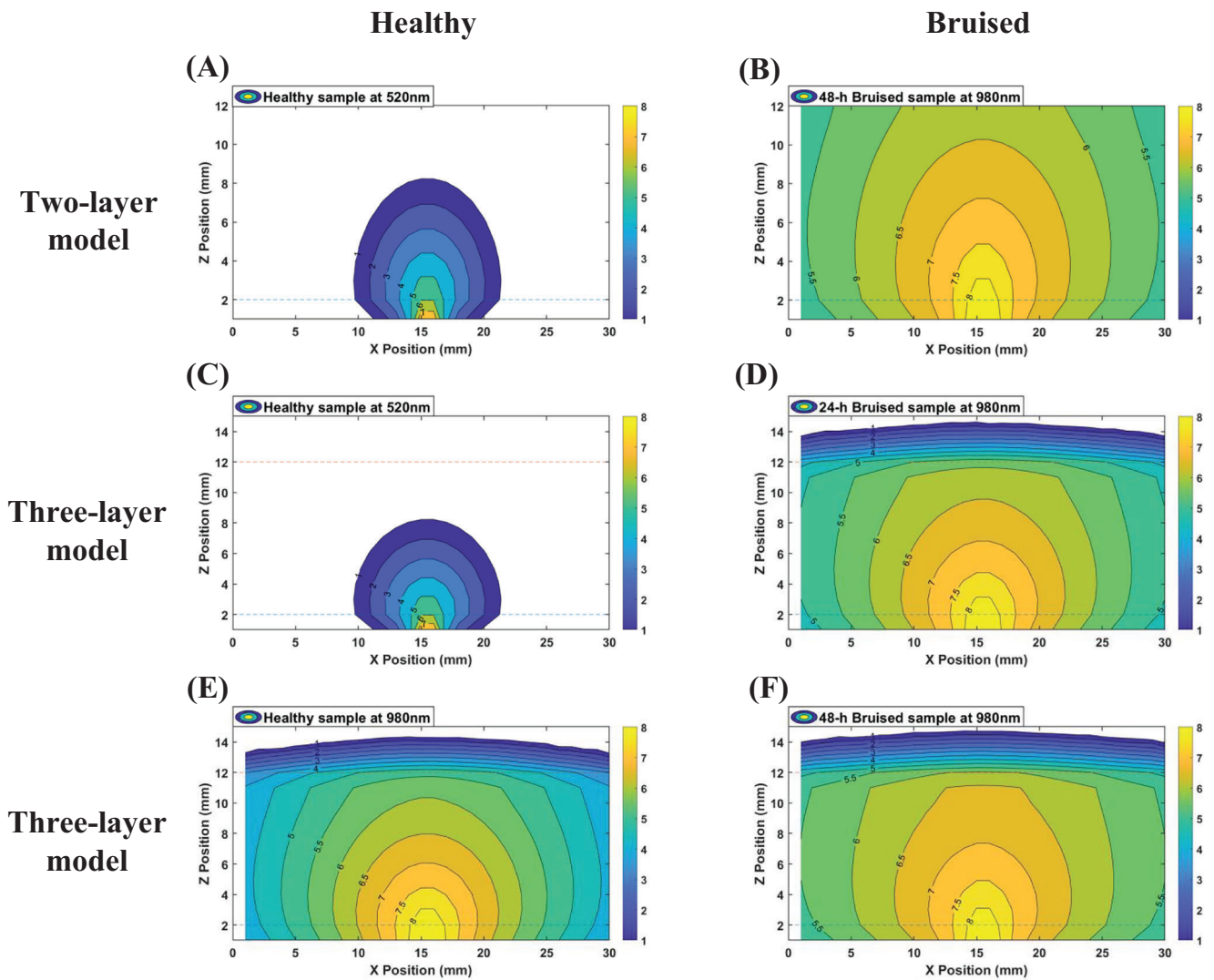


FIGURE 10 | The absorbed energy density for the two-layer model of (A) healthy sample at 520 nm, (B) 48 h bruised sample at 980 nm; the absorbed energy density for the three-layer model of (C) healthy sample at 520 nm, (D) 24 h bruised sample at 980 nm, (E) healthy sample at 980 nm, and (F) 48 h bruised sample at 980 nm.

TABLE 4 | Statistical results on soluble solids content and moisture content of cherry flesh.

Treatment	Quality index	
	Soluble solids content (%)	Moisture content (%)
Healthy	12.63 ± 1.01 ^a	85.04 ± 1.08 ^a
1 h bruised	12.95 ± 1.32 ^b	85.56 ± 1.93 ^b
24 h bruised	13.14 ± 1.51 ^c	86.22 ± 1.53 ^c
48 h bruised	13.23 ± 1.61 ^c	85.74 ± 1.78 ^b

Note: Different letters in the same column indicate they had significant differences ($p \leq 0.05$).

arrangement (Figure 11C). The high anthocyanin content of the pericarp tissue resulted in a larger absorption coefficient for the skin than for the flesh. Skin exhibited significantly greater μ'_s relative to flesh, and the difference originated from the denser and neater cellular structure of the skin tissue.

3.7 | Internal Qualities Prediction Based on OPs

To investigate the relationship between the OPs of cherries and their internal quality content, the results of prediction models for SSC and MC developed based on μ_a and μ'_s in the 400–1050 nm

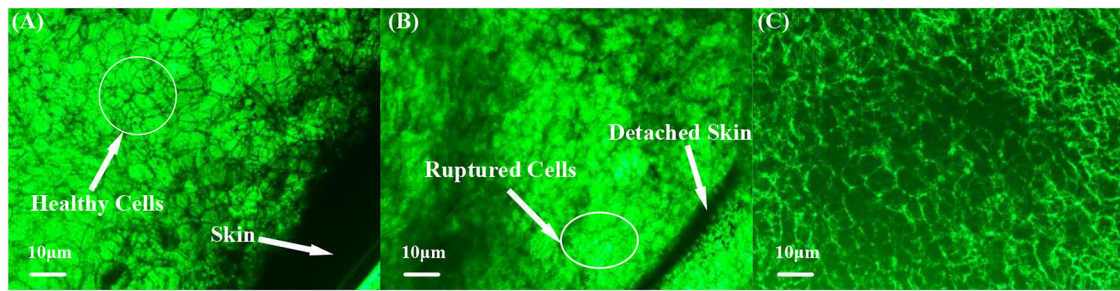


FIGURE 11 | The fluorescence microscopy images (40×magnification) of cherry cells for (A) healthy flesh, (B) 24 h bruised flesh, and (C) skin.

TABLE 5 | Model results of SSC and MC in cherries' flesh based on μ_a and μ'_s .

Parameters	Optical property	Calibration set ($N = 90$)		Prediction set ($N = 30$)	
		R^2_c	RMSEC	R^2_p	RMSEP
SSC	μ_a	0.961	0.198	0.935	0.254
	μ'_s	0.583	0.853	0.457	1.208
MC	μ_a	0.880	0.006	0.839	0.008
	μ'_s	0.289	0.015	0.239	0.019

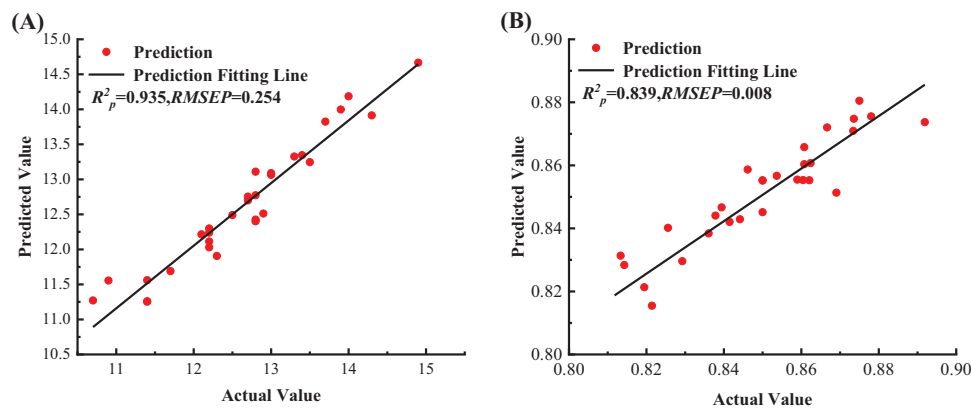


FIGURE 12 | Scatter plots of the prediction set for SSC (A) and MC (B) models based on μ_a in the 400–1050 nm spectral range.

spectral range are presented in Table 5. In predicting SSC and MC, the PLSR model based on μ_a spectroscopy outperformed the μ'_s model (Figure 12), with R^2_p values of 0.935 and 0.839, and RMSEP values of 0.254% and 0.008%, respectively. The high accuracy on the Calibration set ($R^2_c = 0.961$), coupled with a minor decrease in the independent predicting set ($R^2_p = 0.935$), indicates excellent generalization and minimal risk of overfitting. However, the prediction results based on μ'_s are bad and not suitable for detecting the internal components of fruit tissues. These results agree with previous findings by Yang et al. (2024). In addition, Previous studies indicate that simply summing μ_a and μ'_s spectra ($\mu_a + \mu'_s$) does not enhance the predictive performance for SSC or MC (Sun, Lu, Pan et al. 2020; Liu et al. 2021). Therefore, this paper did not consider modeling the combination of the two spectra.

Therefore, the results indicate that stronger correlations were observed between absorption and chemical composition in tissues, suggesting significant potential for predicting cherry SSC and MC using optical properties. Furthermore, when compared

with OPs-based prediction performance reported in other studies, the SSC prediction results in this work were found to be superior to those reported by Wei et al. (2020) for apples ($R^2_p = 0.799$, RMSEP = 0.380%), Fang et al. (2020) for apples ($R^2_p = 0.833$, RMSEP = 0.329%), and Yang et al. (2024) for melons ($R^2_p = 0.86$, RMSEP = 0.79%). However, the MC prediction results were lower than those reported by Yang et al. (2024) for melons ($R^2_p = 0.87$, RMSEP = 0.76%). It should be noted that the variations in prediction results across these studies may be attributed to differences in spectral ranges employed and the specific fruit characteristics analyzed.

4 | Conclusion

This study examined how bruising affects the internal qualities, microstructure, and optical properties of cherry tissues using an integrating sphere system and MCML simulations, as well as the feasibility of internal qualities prediction based on OPs. Healthy

and bruised flesh showed comparable μ_a within 650–1050 nm, but μ'_s was significantly different throughout 400–1050 nm. Compared with healthy flesh, the bruised flesh had a weaker scattering ability but greater light penetration depth, which was mainly related to the incomplete cellular structure of bruised flesh. MCML simulation results have shown that the kernel affected the light propagation properties of cherry fruit tissue, and there is greater potential for detecting cherry bruising and internal quality in the 650–1050 nm range using the reflectance method. However, the reflective mode presents several technical challenges in cherry inspection, such as signal intensity being affected by surface texture and ripeness, environmental light and vibration causing noise interference, and fixed angles resulting in uneven data collection. Future research can further optimize light source power and probe distance, adopt darkroom and vibration-damping designs, and develop multiangle adaptive mechanisms. Understanding how the light propagates through the cherry tissue and determining the light penetration depth, a more precise measuring area can be achieved in the experiment, and the accuracy and robustness of the model can also be improved. However, cherry SSC and MC did not vary significantly throughout the bruising period. To evaluate the predictive utility of OPs, PLSR models were constructed with μ_a and μ'_s to predict SSC and MC in cherries. For predicting SSC and MC, PLSR models using μ_a spectra achieved higher accuracy than those using μ'_s spectra, with R_p^2 of 0.935 and 0.839 and RMSEP of 0.254% and 0.008%, respectively. The results suggested the significant potential for predicting cherry SSC and MC using OPs.

The findings provide a valuable reference for the development of nondestructive testing technologies or instruments for cherry fruits based on VIS-NIR spectroscopy. Although there is progress in the current optical detection study of cherry internal qualities, there are still limitations, such as insufficient consideration of the heterogeneous characteristics of cherry tissues and insufficient consideration of the complex nonlinear relationship between the bruising degree and optical properties. The subsequent research can be carried out in terms of optimizing the light propagation model and establishing quantitative mapping relationships with the help of neural networks and deep learning algorithms, to construct a more efficient and high-precision nondestructive testing method.

Author Contributions

Jian Wu: funding acquisition, investigation, formal analysis, writing – review and editing. **Xianhuan Zeng:** data curation, writing – original draft. **Aiguo Ouyang:** funding acquisition, supervision. **Bin Li:** formal analysis, writing – review and editing. **Nan Chen:** visualization, validation. **Yande Liu:** resources, conceptualization, funding acquisition.

Acknowledgments

This work was supported by the National Key Research and Development Program of China [grant numbers:2022YFD2001804]; Science and Technology Research Program of Jiangxi Provincial Department of Education [grant numbers: GJJ2400503].

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data are available from the corresponding author upon reasonable request.

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